THE RELAXATION TIME IS NOT THE VALUE OF A FLOW PROGRAM

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Abstract. Aldous and Fill ask whether the relaxation time of a reversible Markov chain has a characterization as exactly an infimum over flows. We show that it is not the optimal value of any linear program, or convex quadratic program with affine constraints, whose data are rational functions of the transition probabilities; their congestion and energy bounds are programs of this kind. The optimal value of such a program lies in the field generated by the transition probabilities, while the relaxation time of the simple random walk on the path with six vertices is $3+\sqrt{5}$. Whether a nonlinear functional of flows has optimum exactly the relaxation time remains open. The relaxation time is nevertheless an optimum of exactly the objective that is dual to minimum congestion: the dual of the congestion program maximises over metrics, and replacing metrics by squared Euclidean distance arrays gives twice the relaxation time.

1. Introduction

Let $P=(p_{ij})$ be an irreducible transition matrix on a finite set V with $n=|V|\geq 2$, reversible with respect to its stationary law π, and let $1=\lambda_1>\lambda_2\geq\cdots\geq\lambda_n$ be its eigenvalues. The relaxation time is $\tau_2=1/(1-\lambda_2)$. Write $c_{ij}=\pi_i p_{ij}=\pi_j p_{ji}$ for the conductance of the edge $\{i,j\}$, E for the set of edges $\{i,j\}$ with $i\neq j$ and $c_{ij}>0$ (a loop contributes nothing to the forms below and carries no flow), and

$$\mathcal{E}(f)=\sum_{\{i,j\}\in E} c_{ij}(f_i-f_j)^2, \qquad \mathrm{Var}_\pi(f)=\tfrac{1}{2}\sum_{a,b\in V}\pi_a\pi_b(f_a-f_b)^2,$$

so that $\tau_2=\sup_f \mathrm{Var}_\pi(f)/\mathcal{E}(f)$ over nonconstant f [1, Ch. 3].

Chapter 4 of Aldous and Fill [1] bounds τ_2 above by quantities defined as minima over flows, in the line of Jerrum and Sinclair [8], Diaconis and Stroock [4], Sinclair [13] and the comparison theorems of Diaconis and Saloff-Coste [5]: a family of flows routes a demand $\pi_a\pi_b$ from each a to each b, and a congestion functional of the family, measured against the conductances, is minimised. Their Open Problem 3.44 asks: "Does there exist a characterization of the relaxation time as exactly an inf over flows?" The source adds that "it would be more elegant to derive such inequalities from some exact characterization".

Some restriction on the shape of the characterization is needed for the question to have content, since a constant functional of the flows equal to τ_2 is an infimum over flows. The restriction we impose is the one the chapter's bounds satisfy: a maximum of fixed linear load functionals, or a fixed quadratic energy, minimised over flows in a polyhedral set. Programs of this kind are linear programs, or convex quadratic programs with affine constraints, in the flow variables, with data rational in the entries of P. For them the answer is negative. The question for nonlinear objectives, conic constraints or limits of programs is not settled here.

Theorem 1.1. *Let Φ be a map that assigns to each reversible chain P on a finite graph the optimal value of a program in finitely many real variables that minimises a convex quadratic polynomial (a linear one included) subject to finitely many affine equalities and weak inequalities,*

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whose objective coefficients, objective constant and constraint coefficients are rational functions with rational coefficients of the entries of P, defined at P, and which has a finite optimum. Then $\Phi(P)$ lies in the field $\mathbb{Q}(P)$ generated by the entries of P. For the simple random walk on the path with six vertices, $\mathbb{Q}(P) = \mathbb{Q}$ and $\tau_2 = 3 + \sqrt{5}$. Hence $\Phi(P) \neq \tau_2(P)$ for that chain, and no such Φ equals τ_2 on all reversible chains.

The stationary law is itself in $\mathbb{Q}(P)^V$, being the solution of a nonsingular linear system with coefficients in $\mathbb{Q}(P)$, so coefficients built from π and the conductances c_{ij} are allowed. The path is not special: the simple random walk on the path with five vertices has $\tau_2 = 2 + \sqrt{2}$, on the five-cycle $\tau_2 = 1 + \sqrt{5}/5$, and in a sample of 60 random connected graphs on five to seven vertices with integer edge weights from 1 to 4, the relaxation time of the weighted walk is irrational in 59 (Section 4). Rational values occur when the second eigenvalue is rational, as for K_4 ($\tau_2 = 3/4$) and the Petersen graph ($\tau_2 = 3/2$).

The objective that is dual to minimum congestion does yield τ_2, if metrics are replaced by squared Euclidean distances. For a function d on $V \times V$ put

$$N(d) = \sum_{a,b \in V} \pi_a \pi_b \, d(a,b), \qquad C(d) = \sum_{\{i,j\} \in E} c_{ij} \, d(i,j).$$

Let κ be the minimum, over all families of flows routing $\pi_a \pi_b$ from a to b for every ordered pair $a \neq b$, of the largest ratio of the total load on an edge to its conductance.

Proposition 1.2. (i) *$\kappa = \max\{N(d) : d$ a semimetric on V, $C(d) = 1\}$, where a semimetric is a symmetric nonnegative function on $V \times V$ that vanishes on the diagonal and satisfies the triangle inequality.*

(ii) *$2\tau_2 = \max\{N(d) : d(a,b) = \|x_a - x_b\|^2$ for some $x \colon V \to \mathbb{R}^k$, $k \geq 1$, $C(d) = 1\}$.*

Part (i) is the flow–metric duality for uniform multicommodity flow [10, 11]; part (ii) is the variational formula for τ_2 written for vector-valued functions. The semimetrics in (i) form the metric cone. The functions in (ii), the squared Euclidean distance arrays, are by Schoenberg's theorem [12] the symmetric functions with zero diagonal that are of negative type; we call the cone they form the negative type cone. It is not polyhedral, and its members need not satisfy the triangle inequality. The congestion program and the relaxation time thus optimise the same objective under the same normalisation over two different cones, and the program over the metric cone is linear while the one over the negative type cone is semidefinite. Semidefinite programs over such embeddings appear in the fastest mixing Markov chain problem of Boyd, Diaconis and Xiao [3], whose dual for Markov processes is the maximum variance unfolding problem of Sun, Boyd, Xiao and Diaconis [18], and in the embeddings of Göring, Helmberg and Wappler [7] for the algebraic connectivity. On the path with six vertices the two values are $\kappa = 5$ and $2\tau_2 = 6 + 2\sqrt{5}$.

Theorem 1.1 leaves open whether some natural nonlinear functional of flows has optimum exactly τ_2. The proof applies to any quantity that is irrational on a chain with rational transition probabilities, and in the other direction it says nothing about quantities such as the maximal mean hitting time, which solve linear systems and are rational on rational chains.

2. Programs with data in a subfield

Lemma 2.1. *Let $\mathbb{K} \subseteq \mathbb{R}$ be a subfield. Let Q be a symmetric positive semidefinite $m \times m$ matrix, $q \in \mathbb{K}^m$, $A \in \mathbb{K}^{r \times m}$, $b \in \mathbb{K}^r$, with Q having entries in $\mathbb{K}$, and suppose that $g(x) = \frac{1}{2} x^\top Q x + q^\top x$ is bounded below on the nonempty polyhedron $F = \{x \in \mathbb{R}^m : Ax \leq b\}$. Then g attains its infimum on F at a point of $\mathbb{K}^m$, and the infimum lies in $\mathbb{K}$.*

Equality constraints are included as pairs of inequalities, and $Q = 0$ gives linear programs.

Proof. A quadratic function bounded below on a nonempty polyhedron attains its infimum there [6]; let x^* be a minimiser and I the set of rows of A active at x^*. Since g is convex and F is polyhedral, the Karush–Kuhn–Tucker conditions hold at x^* with some $y^* \geq 0$, and conversely any (x, y) satisfying them makes x a minimiser. Consider the polyhedron

$$T = \{(x, y) \in \mathbb{R}^m \times \mathbb{R}^I : \ Qx + q + A_I^\top y = 0,\ A_I x = b_I,\ y \geq 0,\ Ax \leq b\}.$$

It is defined by linear equalities and inequalities with coefficients in $\mathbb{K}$ and contains (x^*, y^*). A nonempty polyhedron has a minimal nonempty face G. Such a face is an affine subspace: if some defining inequality were tight at one point of G and slack at another, making it tight would give a smaller nonempty face. Hence G is exactly the solution set of the system of linear equations obtained from the defining equalities of T together with the inequalities tight on G, and every solution of that system lies in T. The system has coefficients in $\mathbb{K}$ and a real solution, so by Gaussian elimination over $\mathbb{K}$ it has a solution in $\mathbb{K}^{m+|I|}$. That point $(\bar{x}, \bar{y})$ lies in T, so $\bar{x}$ satisfies the Karush–Kuhn–Tucker conditions and minimises g on F, and $g(\bar{x}) \in \mathbb{K}$. □

Proof of Theorem 1.1. Fix a chain P and set $\mathbb{K} = \mathbb{Q}(P) \subseteq \mathbb{R}$. The program defining $\Phi(P)$ has data in $\mathbb{K}$ and a finite optimum, so $\Phi(P) \in \mathbb{K}$ by Lemma 2.1. For the simple random walk on the path $0, 1, \ldots, 5$ every nonzero transition probability is $1/2$ or 1, so $\mathbb{K} = \mathbb{Q}$. The eigenvalues of this chain are $\cos(\pi k/5)$, $k = 0, \ldots, 5$, as one checks with the eigenvectors $f_j = \cos(\pi kj/5)$. Hence $\lambda_2 = \cos(\pi/5) = (1 + \sqrt{5})/4$ and $\tau_2 = 4/(3 - \sqrt{5}) = 3 + \sqrt{5}$, which is irrational. □

The same eigenvectors give $\lambda_2 = \cos(\pi/4)$ and $\tau_2 = 2 + \sqrt{2}$ on the path with five vertices; on the five-cycle $\lambda_2 = \cos(2\pi/5) = (\sqrt{5} - 1)/4$ and $\tau_2 = 1 + \sqrt{5}/5$.

Remark 2.2. The two flow functionals that appear in the bounds are covered when their coefficients are fixed. A congestion functional with fixed loads is a maximum of linear forms in the flows and is minimised by a linear program after introducing one more variable. An energy $\sum_e f_e^2/c_e$ of a unit flow is a convex quadratic function, and its minimum, the effective resistance, is rational on rational chains. On the path with six vertices the end-to-end energy is 50 and $\kappa = 5$.

3. Two cones

Proof of Proposition 1.2. (i) The congestion program is the linear program of minimising t over flows $f^{ab} \geq 0$ on the directed edges, one for each ordered pair $a \neq b$, each sending $\pi_a \pi_b$ from a to b, subject to $\sum_{a \neq b} (f_{ij}^{ab} + f_{ji}^{ab}) \leq t\, c_{ij}$ for every edge. Its dual is to maximise $\sum_{a \neq b} \pi_a \pi_b\, \delta_\ell(a, b)$ over edge lengths $\ell \geq 0$ with $\sum_E c_{ij} \ell_{ij} = 1$, where δ_ℓ is the shortest path distance for the lengths ℓ [10, 11]. Write M for the maximum in (i). The function δ_ℓ is a semimetric with $\delta_\ell(i, j) \leq \ell_{ij}$ on every edge, so $C(\delta_\ell) \leq C(\ell) = 1$; the inequality can be strict, as on the triangle with $\pi_i = 1/3$, $c_e = 1/6$ and lengths $(4, 1, 1)$, where $C(\delta_\ell) = 2/3$. If $C(\delta_\ell) > 0$, then $\delta_\ell / C(\delta_\ell)$ is feasible for M and $N(\delta_\ell) \leq N(\delta_\ell)/C(\delta_\ell) \leq M$. If $C(\delta_\ell) = 0$, then δ_ℓ vanishes on every edge, hence everywhere since the graph is connected, and $N(\delta_\ell) = 0 \leq M$. Conversely, a semimetric d with $C(d) = 1$ defines lengths $\ell_{ij} = d(i, j)$ with the same normalisation, and $\delta_\ell \geq d$ by the triangle inequality, so $N(\delta_\ell) \geq N(d)$. Both maxima are attained: the set of normalised lengths is compact, and so is the set of normalised semimetrics, since $d(i, j) \leq 1/c_{ij}$ on edges and the triangle inequality along a fixed path bounds every other value. The two maxima therefore coincide, and by duality they equal κ.

(ii) For $x \colon V \to \mathbb{R}^k$ with coordinates $x^{(1)}, \ldots, x^{(k)}$ and $d(a, b) = \|x_a - x_b\|^2$,

$$N(d) = 2 \sum_{s=1}^{k} \mathrm{Var}_\pi(x^{(s)}) \leq 2\tau_2 \sum_{s=1}^{k} \mathcal{E}(x^{(s)}) = 2\tau_2\, C(d),$$

by the variational formula applied to each coordinate. Equality holds for $k = 1$ and x an eigenvector of λ_2, normalised so that $C(d) = 1$. □

The negative type cone on V is not polyhedral when $|V| \geq 3$, since fixing one point at the origin identifies it linearly with the cone of positive semidefinite $(|V|-1) \times (|V|-1)$ Gram matrices. Squared Euclidean distance arrays need not satisfy the triangle inequality (for the points $0, 1, 2$ of the line, $4 > 1 + 1$), and metrics need not be of negative type (the graph metric of $K_{2,3}$ is not), so neither cone contains the other. The semidefinite relaxation of sparsest cut of Arora, Rao and Vazirani [2] optimises over their intersection, the squared Euclidean distance arrays that also satisfy the triangle inequality. Kahale [9] showed that the best bound on the spectral gap given by flows with weighted paths is the value of a semidefinite program and differs from the spectral gap by at most a polylogarithmic factor in the number of states. Proposition 1.2 places κ and $2\tau_2$ as values of one objective on two different cones. It does not give an inequality between them, and none holds in general from this description alone.

4. Computations

The program `code/checks.py` in the repository [14] computes λ_2 and τ_2 exactly from the characteristic polynomial of the transition matrix for the simple random walks on P_5, P_6, C_5, K_4 and the Petersen graph, compares them with the closed forms above, and reads off irrationality from the minimal polynomial of τ_2 ($x^2 - 6x + 4$ for P_6). It computes κ and the end-to-end energy exactly on the two paths (15/4 and 32 on P_5, 5 and 50 on P_6), and solves the congestion program on C_5 as a linear program, with value 12/5. It checks Proposition 1.2(ii) by evaluating $N(d)/C(d)$ at an eigenvector of λ_2, which returns $2\tau_2$, and at 4,000 random configurations in dimensions 1 to 4 for each chain, none of which exceeds it. Finally, for 60 random connected graphs on five to seven vertices with integer edge weights between 1 and 4, it finds τ_2 irrational in 59 cases.

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This work was done by multiple AI agents, Claude (Anthropic) and Codex (OpenAI), connected to Math Brain [15] and working together inside the Brain framework [17], a shared, persistent memory in which the agents record claims, proofs, failed attempts and methods, retrieve one another's results, coordinate who works on what, and audit each other's work. For more on the details behind the Brain and Math Brain, see the Math Brain paper [16], the Math Brain repository [15] and the Brain framework repository [17].

Declaration of competing interest

The author holds equity in Heights Labs and in Convex Nexus Capital and has a management role at Convex Nexus Capital. These entities are included in the funding declaration above.

Declaration of generative AI and AI-assisted technologies

The paper comes out of Math Brain, an ongoing project on the use of AI systems for research mathematics. In September 2026 Claude models (Anthropic), running as research agents of that project, located Open Problem 3.44 in the source, proposed the rationality argument for linear programs with the path witness, and computed the flow values. In November 2026 Claude Opus 5.5 (Anthropic), working in Claude Code, checked the argument, extended it to convex quadratic programs through Lemma 2.1, proved Proposition 1.2, wrote the verification program and wrote the paper. The author directed the work and is responsible for the content of the paper and for decisions concerning its dissemination.

Data availability

No data were used. The code is available at `https://github.com/yspennstate/relaxation-time-flow-programs`.

The paper is part of the Math Brain project, whose public repository is `https://github.com/yspennstate/math-brain-public` [15].

Appendix A. Notation

Symbols are listed in the order in which they first appear.

Symbol	Meaning	First appears
$P = (p_{ij})$	irreducible reversible transition matrix	Section 1
V, n	finite state space, $n = \lvert V\rvert \geq 2$	Section 1
π	stationary law of P	Section 1
$\lambda_1, \dots, \lambda_n$	eigenvalues of P, $1 = \lambda_1 > \lambda_2 \geq \cdots \geq \lambda_n$	Section 1
τ_2	relaxation time $1/(1-\lambda_2)$	Section 1
c_{ij}	conductance $\pi_i p_{ij}$ of the edge $\{i, j\}$	Section 1
E	edges $\{i, j\}$, $i \neq j$, with $c_{ij} > 0$	Section 1
$\mathcal{E}(f)$	Dirichlet form $\sum_E c_{ij}(f_i - f_j)^2$	Section 1
$\mathrm{Var}_\pi(f)$	variance of f under π	Section 1
Φ	optimal value of a linear or convex quadratic program with data in $\mathbb{Q}(P)$	Theorem 1.1
$\mathbb{Q}(P)$	field generated over $\mathbb{Q}$ by the entries of P	Theorem 1.1
$N(d)$	$\sum_{a,b} \pi_a \pi_b \, d(a, b)$	Section 1
$C(d)$	$\sum_E c_{ij} \, d(i, j)$	Section 1
κ	minimum congestion of uniform multicommodity flow	Section 1
$x \colon V \to \mathbb{R}^k$	embedding defining a squared Euclidean distance array	Proposition 1.2
metric cone	semimetrics on V	Section 1
negative type cone	squared Euclidean distance arrays on V	Section 1
$\mathbb{K}$	subfield of $\mathbb{R}$ containing the data	Lemma 2.1
Q, q, g	quadratic objective $g(x) = \frac{1}{2} x^\top Q x + q^\top x$	Lemma 2.1
A, b, F	constraints and feasible polyhedron $F = \{Ax \leq b\}$	Lemma 2.1
x^*, y^*	minimiser of g on F and its multipliers	Section 2
I	rows of A active at x^*	Section 2
T	polyhedron of Karush–Kuhn–Tucker pairs	Section 2
G	minimal nonempty face of T	Section 2
ℓ, δ_ℓ	edge lengths and their shortest path distance	Section 3
M	maximum in Proposition 1.2(i)	Section 3

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